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Astronomy (AEC Course) Level – 2 Report on

“Exoplanet Transit Detection using Box Least Squares (BLS) Algorithm”

Submitted in Partial Fulfillment of the Requirements for the Course

Astronomy-Explore the Space

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DEPARTMENT OF AEROSPACE ENGINEERING

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CERTIFICATE

Certified that the project work titled "**Exoplanet Transit Detection using Box Least Squares (BLS) Algorithm**" is carried out by **GN Prajin (1MJ24IS022)**, **Likith Gowda (1MJ24IS042)**, **Shaik Mohammed Afsar (1MJ24AE076)** and **Balaram (1MJ24AE015)** are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of Bachelor of Engineering during the year 2025 - 2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution for the said Degree.

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DECLARATION

We, **GN Prajin (1MJ24IS022), Likith Gowda (1MJ24IS042), Shaik Mohammed Afsar (1MJ24AE076) and Balaram (1MJ24AE015)**, the students of Fourth semester B.E., MVJ College of Engineering, Bengaluru, hereby declare that the **ASTRONOMY EXPLORE THE SPACE - LEVEL 02** titled '**Exoplanet Transit Detection using Box Least Squares (BLS) Algorithm**' has been carried out by us and submitted in partial fulfilment for the award of the Degree of Bachelor of Engineering during the year 2025-2026.

Further, We declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any intellectual property rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru, and I will be one of the authors of the same.

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ABSTRACT

The discovery of planets beyond our Solar System, known as exoplanets, has become one of the most important fields in modern astronomy. Among the various exoplanet detection techniques, the transit method has proven to be highly successful due to its ability to identify periodic reductions in stellar brightness caused by a planet passing in front of its host star. Detecting these transit signals is challenging because the changes in brightness are often extremely small and are frequently masked by noise, stellar variability, and observational errors.

This project focuses on the study and implementation of the Box Least Squares (BLS) algorithm, a widely used technique for detecting periodic transit signals in astronomical light curves. The BLS algorithm searches for box-shaped dips in stellar brightness by fitting a periodic box model over a range of trial orbital periods and transit durations. The method is particularly effective because planetary transits typically produce short-duration decreases in brightness compared to the overall orbital period.

A detailed literature review was conducted on foundational and modern research related to BLS and exoplanet transit detection. The review covered the original BLS algorithm, its practical implementation through the Astropy library, computational improvements through fast-folding approaches, modern alternatives such as Transit Least Squares (TLS), and comparative studies involving different periodogram algorithms.

The proposed methodology involves acquiring real astronomical data from NASA Kepler and TESS missions, preprocessing the light curves, applying the BLS algorithm, generating periodograms, and identifying potential transit candidates. The study establishes a strong foundation for future implementation and evaluation of exoplanet detection techniques using real-world astronomical datasets.

PREAMBLE

Astronomy has always played a crucial role in understanding the universe and humanity's place within it. One of the most exciting developments in recent decades has been the discovery of exoplanets, which are planets orbiting stars outside our Solar System. The search for exoplanets not only expands our knowledge of planetary systems but also contributes to the broader quest for potentially habitable worlds.

Advancements in observational astronomy have enabled large-scale missions such as NASA's Kepler and Transiting Exoplanet Survey Satellite (TESS) to collect enormous amounts of photometric data. These missions continuously monitor the brightness of stars and generate light curves that can reveal the presence of planets through periodic transit events.

The analysis of such large datasets requires sophisticated computational techniques capable of distinguishing genuine transit signals from noise and stellar variability. The Box Least Squares (BLS) algorithm is one of the most widely used methods for this purpose. Since its introduction in 2002, BLS has become a standard tool in exoplanet transit detection pipelines due to its effectiveness in identifying periodic box-shaped brightness reductions.

This report presents a comprehensive study of the BLS algorithm, its theoretical foundations, literature background, methodology, implementation approach, challenges, and expected outcomes. The work serves as a preparatory stage for implementing the algorithm on real astronomical datasets and evaluating its performance in identifying exoplanet transit candidates.

ACRONYMS

ACRONYM	FULL FORM
AEC	Ability Enhancement Course
VTU	Visvesvaraya Technological University
BLS	Box Least Squares
TLS	Transit Least Squares
TESS	Transiting Exoplanet Survey Satellite
NASA	National Aeronautics and Space Administration
SR	Signal Residue
SDE	Signal Detection Efficiency

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CHAPTER 01

Introduction

CHAPTER 01

INTRODUCTION

1.1 Exoplanets

Exoplanets are planets that orbit stars other than the Sun. Since the first confirmed exoplanet discovery in 1992, thousands of exoplanets have been identified using various observational methods. These discoveries have transformed our understanding of planetary formation and the diversity of planetary systems in the universe.

Exoplanets vary significantly in size, composition, and orbital characteristics. They include gas giants larger than Jupiter, rocky Earth-like planets, and planets that orbit extremely close to or far from their host stars.

1.2 Transit Method

The transit method detects exoplanets by observing the periodic reduction in a star's brightness when a planet passes between the star and the observer.

When a transit occurs:

- **A small fraction of the star's light is blocked.**
- **The observed brightness decreases temporarily.**
- **The event repeats periodically according to the planet's orbital period.**

The depth, duration, and frequency of these brightness dips provide information about the planet's size and orbit.

1.3 Light Curves

A light curve is a graph showing stellar brightness as a function of time.

Light curves are fundamental to transit detection because they contain the observable signatures of planetary transits. However, they often contain noise from:

- Instrumentation errors
- Stellar activity
- Data gaps
- Measurement uncertainties

Consequently, specialized algorithms are required to identify transit signals accurately.

1.4 Need for Transit Detection Algorithms

Since transit signals are often weak and hidden within noisy observations, automated computational methods are essential. Transit detection algorithms search for periodic patterns within large datasets and identify potential exoplanet candidates.

The Box Least Squares algorithm is one of the most successful methods developed specifically for this purpose.

CHAPTER 02

Objectives of the project

CHAPTER 02

Objectives of the project

The primary objectives of this project are:

1. To study the principles of exoplanet detection using transit photometry.
2. To understand the mathematical and computational foundations of the Box Least Squares algorithm.
3. To analyze concepts such as phase folding, transit duration estimation, Signal Residue (SR), and Signal Detection Efficiency (SDE).
4. To investigate the implementation of BLS using Python and the Astropy BoxLeastSquares module.
5. To study the application of BLS on real astronomical datasets obtained from NASA missions.
6. To generate and interpret BLS periodograms.
7. To compare BLS with advanced alternatives such as Transit Least Squares (TLS).
8. To identify current limitations and future research opportunities in exoplanet transit detection.

CHAPTER 03

Literature Survey

CHAPTER 03

LITERATURE SURVEY

3.1 Literature Review

This section reviews key research papers relevant to exoplanet detection using Kepler data and the BLS algorithm.

1. A Box-Fitting Algorithm in the Search for Periodic Transits (2002)

Authors: Géza Kovács, Shay Zucker and Tsevi Mazeh

This landmark paper introduced the Box Least Squares algorithm. The authors proposed fitting a box-shaped model to stellar light curves and searching across many trial periods.

Major Contributions:

- *Introduced the BLS algorithm.*
- *Demonstrated superior performance compared to Fourier methods for transit detection.*
- *Established a statistical framework for evaluating transit candidates.*
- *Defined Signal Residue (SR) as a measure of transit significance.*

The study became the foundation of modern transit detection pipelines.

Inference

The paper proved that transit signals are better represented using box-shaped models rather than sinusoidal functions, making BLS highly suitable for exoplanet discovery.

2. BoxLeastSquares Documentation (Astropy)

Author: Astropy Collaboration

The Astropy implementation provides a practical framework for applying BLS in Python.

Key Features:

- Efficient period grid generation.
- Transit duration estimation.
- Periodogram generation.
- Statistical evaluation of transit candidates.
- Easy integration with astronomical datasets.

Inference

The documentation serves as a bridge between theoretical understanding and practical implementation of BLS.

3. Transit Least Squares (TLS) Algorithm (2019)

Authors: Michael Hippke and René Heller

TLS was developed to address limitations of BLS.

Key Contributions:

- Uses realistic transit shapes.
- Models stellar limb darkening.
- Improves detection sensitivity.
- Performs better for Earth-sized planets.

Inference

TLS demonstrates that physically realistic transit models can improve detection accuracy for shallow transit signals.

4. fBLS – Fast Folding BLS Algorithm (2022)

Authors: Shahaf, Zackay, Mazeh, Faigler and Ivashtenko

This work focused on improving computational efficiency.

Key Contributions:

- Reduced execution time.
- Optimized period searches.
- Improved scalability for large surveys.

Inference

Computational performance is becoming increasingly important as astronomical datasets continue to grow.

5. A Study of Two Periodogram Algorithms for Exoplanet Detection (2023)

Authors: Yash Gondhalekar et al.

This paper compared BLS and Transit Comb Filter (TCF).

Major Findings:

- BLS performs strongly for box-shaped signals.
- TCF may outperform BLS in specific scenarios.
- Algorithm selection depends on observational conditions.

Inference

No single transit detection algorithm is universally optimal, emphasizing the need for comparative evaluation.

3.2 Literature Gap

The literature review reveals several research gaps and opportunities for improvement.

1. The traditional BLS algorithm assumes a perfect box-shaped transit, whereas real planetary transits exhibit curved profiles due to stellar limb darkening.
2. The computational cost of searching thousands of trial periods remains significant for large astronomical surveys.
3. Noise and stellar variability continue to reduce detection accuracy, especially for small exoplanets.
4. Existing studies primarily focus on algorithm development, while fewer studies investigate practical implementation using publicly available datasets.
5. Comparative studies between BLS and modern algorithms such as TLS remain limited.

This project aims to address part of this gap by studying, implementing, and evaluating BLS using real Kepler and TESS light curve datasets.

3.3 Methodology

1. Data Acquisition

Astronomical light curve data will be obtained from:

- NASA Kepler Mission
- NASA TESS Mission

These datasets contain brightness measurements collected over extended periods.

2. Data Preprocessing

Preprocessing steps include:

- Removal of missing values
- Outlier detection
- Noise reduction
- Flux normalization

The objective is to prepare clean and consistent data for transit detection.

3. Phase Folding

Phase folding converts observation times into orbital phases.

Benefits:

- Aligns repeated transit events.
- Enhances visibility of periodic signals.
- Improves transit detection accuracy.

4. Application of BLS

The BLS algorithm searches for:

- Trial periods
- Transit durations
- Transit depths

5. Periodogram Generation

The algorithm generates a periodogram showing detection strength versus trial period.

Peaks indicate likely orbital periods.

6. Candidate Identification

The strongest peaks are analyzed further to identify potential exoplanet transit candidates.

CHAPTER 04

BOX LEAST SQUARES ALGORITHM

CHAPTER 04

BOX LEAST SQUARES ALGORITHM

4.1 Principle

The Box Least Squares (BLS) algorithm is a specialized transit detection method used to identify periodic decreases in stellar brightness caused by exoplanets passing in front of their host stars. Unlike traditional Fourier-based techniques that are designed to detect sinusoidal signals, BLS is specifically optimized for identifying box-shaped transit signals commonly observed in astronomical light curves. The algorithm operates by testing a large number of trial orbital periods and fitting a box-shaped model to the folded light curve at each period. Since planetary transits typically produce short-duration dips compared to the overall orbital period, a box-shaped approximation provides an effective way to identify potential transit events. The period that produces the strongest and most statistically significant signal is considered a potential exoplanet candidate.

4.2 Working procedure

The working procedure of the BLS algorithm begins with a preprocessed light curve containing brightness measurements recorded over time. A range of possible orbital periods is selected, and the light curve is phase folded for each trial period. Phase folding aligns repeated transit events, making periodic signals easier to identify. After folding the data, the algorithm fits a box-shaped transit model characterized by parameters such as transit depth, duration, and phase. The quality of the fit is then evaluated using statistical measures that quantify how well the model explains the observed data. This process is repeated for all trial periods and transit durations within the search range. The results are combined to generate a periodogram, where peaks indicate periods that are likely to correspond to genuine transit signals. The period associated with the highest peak is generally selected as the most probable orbital period of the detected exoplanet candidate.

4.3 Signal Residue (SR)

Signal Residue (SR) is one of the statistical measures used within the BLS algorithm to evaluate the significance of a potential transit signal. It represents the difference between the observed light curve and the fitted transit model and provides an indication of how strongly the data supports the presence of a transit event. A larger Signal Residue value generally corresponds to a stronger transit-like signal within the light curve. During the period search process, SR values are calculated for different trial

periods and transit durations. These values help identify regions where the box-shaped model fits the data particularly well. The concept of Signal Residue plays an important role in distinguishing genuine transit signals from random noise and other brightness variations present in astronomical observations.

4.4 Signal Detection Efficiency (SDE)

Signal Detection Efficiency (SDE) is a statistical parameter used to assess the significance of peaks observed in a BLS periodogram. While Signal Residue measures the strength of an individual transit signal, SDE evaluates that signal relative to the background noise distribution. It provides a normalized measure of detection confidence by comparing the strongest periodogram peak with the average and standard deviation of the surrounding periodogram values. A high SDE value indicates that the detected signal stands out significantly from the noise and is therefore more likely to represent a genuine exoplanet transit. Conversely, low SDE values suggest that the observed peak may be caused by random fluctuations or systematic noise. As a result, SDE is widely used as a criterion for selecting promising transit candidates for further investigation.

CHAPTER 05
Implementation

CHAPTER 05

Implementation

The implementation of the Box Least Squares algorithm will be carried out using Python due to its extensive scientific computing capabilities and the availability of specialized astronomical libraries. The primary tools used in this project include **NumPy** for numerical computations, **Matplotlib** for visualization, and **Astropy** for transit detection and periodogram generation. Before working with real astronomical datasets, a simulated light curve is created to demonstrate the fundamental principles of transit detection. The simulated data contains periodic reductions in brightness that mimic the effect of a planet transiting its host star.

After generating the simulated light curve, the `BoxLeastSquares` module from the `Astropy` library is applied to search for periodic transit signals. A range of trial orbital periods is specified, and the algorithm evaluates each period by fitting a box-shaped transit model. The resulting BLS periodogram displays the detection strength as a function of period, allowing the most significant transit signal to be identified. The period corresponding to the highest peak in the periodogram is selected as the best estimate of the orbital period.

In the next phase of the project, the same implementation procedure will be extended to real astronomical light curves obtained from NASA's Kepler and TESS missions. Additional preprocessing steps such as handling missing observations, removing noise, and normalizing brightness measurements will be performed before applying the BLS algorithm. The final outputs will include light curve visualizations, periodograms, phase-folded transit curves, and the identification of potential exoplanet transit candidates.

Code:

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  time = np.linspace(0, 30, 3000)
5
6  flux = np.ones_like(time)
7
8  period = 5
9  duration = 0.2
10 depth = 0.02
11
12 for i, t in enumerate(time):
13     phase = t % period
14     if phase < duration:
15         flux[i] -= depth
16
17 flux += np.random.normal(0, 0.003, len(flux))
18
19 plt.plot(time, flux, '.')
20 plt.show()
```

Output:

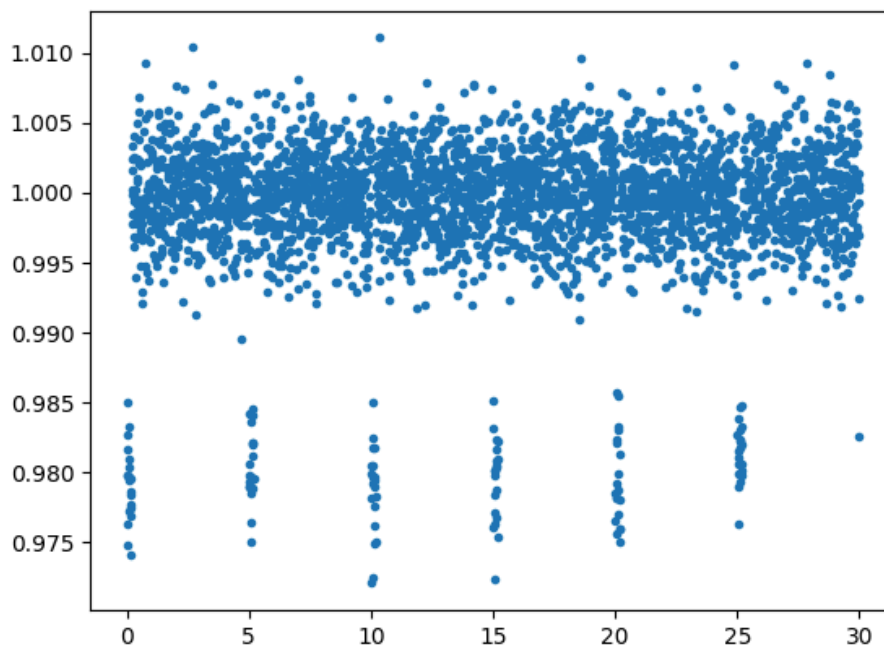


Fig. 5.1 Simulated Light curve

Code:

```
26 from astropy.timeseries import BoxLeastSquares
27
28 model = BoxLeastSquares(time, flux)
29
30 periods = np.linspace(1, 10, 5000)
31
32 results = model.power(
33     periods,
34     0.2
35 )
36
37 best_period = results.period[np.argmax(results.power)]
38
39 print(best_period)
40
41 plt.plot(results.period,
42          results.power)
43
44 plt.xlabel("Period")
45 plt.ylabel("BLS Power")
46 plt.show()
```

Output:

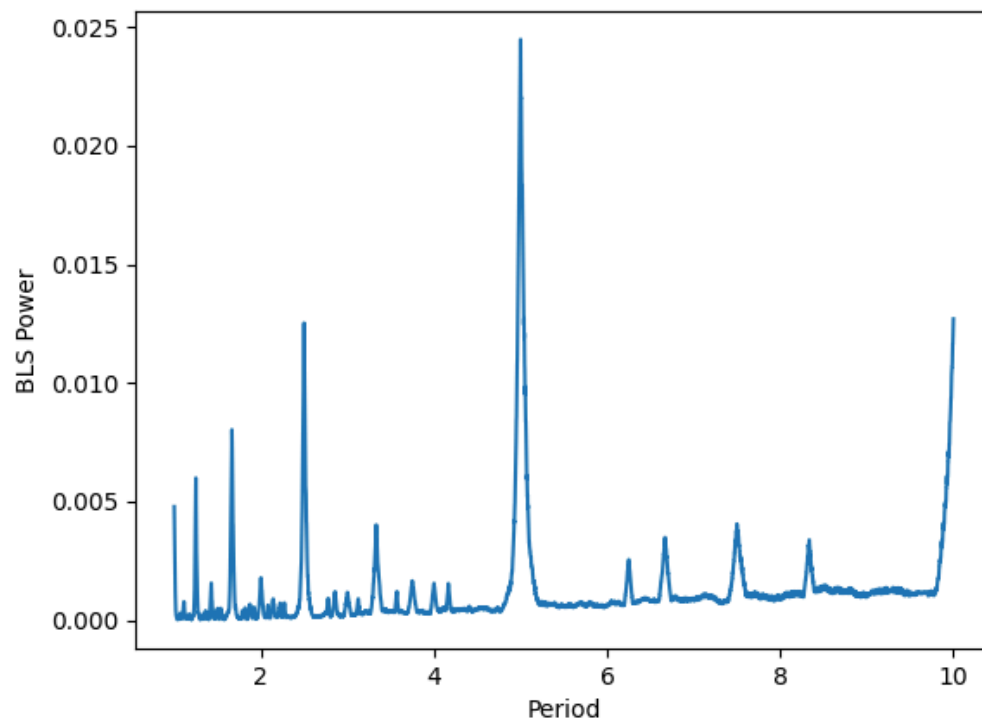


Fig. 5.2 Periodogram from the BLS algorithm

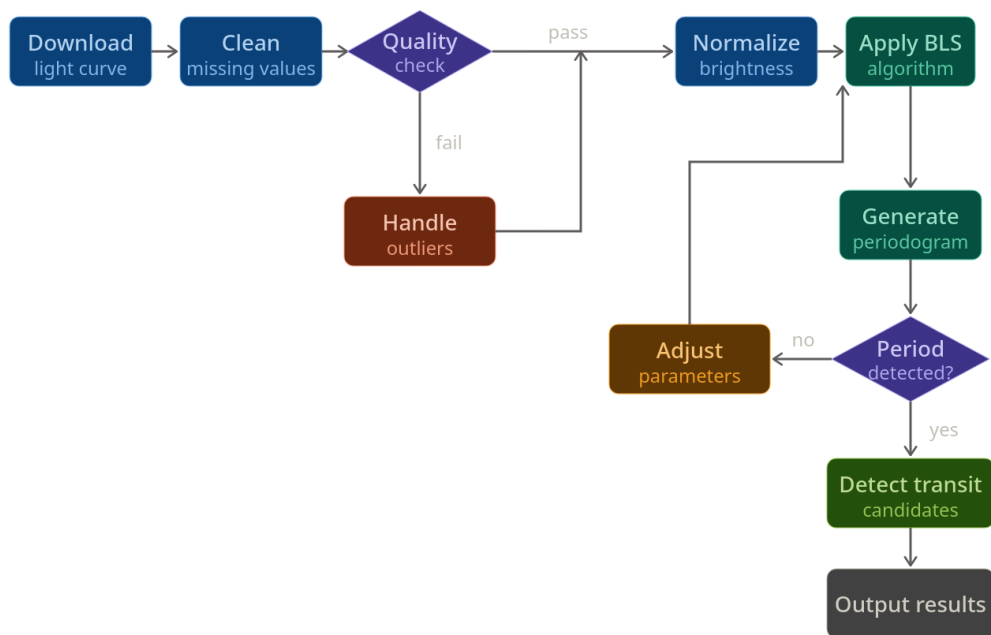


Fig. 5.3 Implementation Flow Chart

CHAPTER 06

Conclusion

CHAPTER 06

Conclusion

6.1 Results and Discussion

The work completed during this phase of the project provided a comprehensive understanding of the Box Least Squares algorithm and its role in exoplanet transit detection. Through an extensive review of relevant literature, the theoretical foundations of transit photometry, phase folding, Signal Residue, and Signal Detection Efficiency were studied in detail. The literature review highlighted the effectiveness of BLS in detecting periodic transit signals while also revealing limitations related to computational complexity and the simplified box-shaped transit model.

The study also examined modern developments in transit detection, including Transit Least Squares and optimized variants such as fBLS. These investigations demonstrated how researchers continue to improve detection sensitivity and computational efficiency in response to the growing volume of astronomical data. Simulated light curve analysis further helped illustrate the practical application of the algorithm and the interpretation of BLS periodograms. Overall, the findings establish a strong theoretical foundation for the implementation and evaluation stages of the project.

6.2 Level – 3 Plan of Work

The next phase of the project will focus on implementing the Box Least Squares algorithm on real astronomical datasets. Light curve observations from NASA's Kepler and TESS missions will be collected and preprocessed to remove noise, handle missing values, and normalize brightness measurements. Once the datasets have been prepared, the Astropy BoxLeastSquares implementation will be used to search for periodic transit signals over a selected range of orbital periods and transit durations.

The generated periodograms will be analyzed to identify significant peaks corresponding to potential exoplanet candidates. Additional visualizations, including folded light curves, will be created to verify the periodic nature of detected signals. The obtained results will then be compared with published exoplanet parameters to evaluate detection accuracy. The performance of the BLS algorithm will also be assessed in terms of sensitivity, computational efficiency, and robustness against noise. This work will provide practical experience with real astronomical observations and enable a detailed evaluation of the algorithm's effectiveness.

6.3 Conclusion

This study successfully established the theoretical and methodological foundation required for exoplanet transit detection using the Box Least Squares algorithm. The principles of transit photometry, light curve analysis, phase folding, and statistical transit detection were investigated through a detailed review of contemporary research literature. Particular attention was given to understanding how the BLS algorithm identifies periodic transit signals and why it remains one of the most widely used techniques in modern exoplanet discovery.

The literature review revealed both the strengths and limitations of existing approaches while identifying opportunities for further improvement. A clear implementation strategy using Python and the Astropy BoxLeastSquares module was developed, and preliminary testing using simulated light curves demonstrated the fundamental operation of the algorithm. The knowledge and methodology acquired during this phase provide a solid foundation for future implementation using real Kepler and TESS observations. The subsequent phase of the project will focus on practical application, performance evaluation, and validation of detected transit candidates.

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